ASEAN GEOAI FUSION 2026

Universiti Kebangsaan Malaysia
The National University of Malaysia

AGAIF 2026 BOOTCAMP 1: FOUNDATIONAL GEOAI

What is GeoAI?
Session 5

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WHAT IS GeoAI?

Artificial Intelligence for Geospatial Intelligence



Integrating **Artificial Intelligence**, **GIS, Remote Sensing** and **Big Spatial Data** for Smarter Decision-Making

GIS
Spatial data management, analysis & visualization

+

REMOTE SENSING
Earth observation from satellites, drones and sensors

+

ARTIFICIAL INTELLIGENCE
Machine learning, deep learning, pattern recognition

+

BIG SPATIAL DATA
Large volume, variety and velocity of location-based data

LEARNING OUTCOMES

At the end of this lecture, students will be able to:

01 Define **GeoAI** and explain its scope.

02 Describe the **technologies** behind **GeoAI**.

03 Explain GeoAI applications in the **telecommunications** industry.

04 Discuss GeoAI challenges and opportunities in **ASEAN**.

05 Analyse real-world GeoAI **case studies**.

GeoAI empowers us to **understand the world spatially**, predict what's next, and make **better decisions** for a sustainable future.

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WHAT IS GeoAI?



DEFINITION

GeoAI (Geospatial Artificial Intelligence) is an interdisciplinary field that combines

ARTIFICIAL INTELLIGENCE (AI)
Enables machines to simulate human intelligence and solve complex problems.

GEOGRAPHIC INFORMATION SYSTEMS (GIS)
Manages, analyzes and visualizes spatial data.

REMOTE SENSING
Captures information about the Earth's surface using satellites, drones and sensors.

SPATIAL STATISTICS
Analyzes spatial data to detect patterns and relationships.

MACHINE LEARNING
Allows systems to learn from data and improve predictions.

DEEP LEARNING
Uses neural networks to extract complex patterns from large datasets.

BIG GEOSPATIAL DATA
Handles large volume, variety and velocity of spatial data.

to automatically extract knowledge, detect patterns, make predictions, and support spatial decision-making.



SIMPLE DEFINITION

AI + GIS + SPATIAL DATA + BIG DATA ANALYTICS = **GeoAI**

The power of AI applied to spatial data to understand the world and make smarter decisions.

GeoAI is transforming how we: Monitor our planet | Manage resources | Plan smart cities | Respond to disasters | Support sustainability

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EVOLUTION OF GeoAI

From Spatial Data to Intelligent Decisions for a Smarter Future

GeoAI represents the next evolution of geospatial technology, powered by AI and big data, driving smarter insights and sustainable solutions.



01 **TRADITIONAL GIS** — Early GIS focused on mapping and visualizing spatial data.

02 **SPATIAL DATABASE** — Structured storage and management of spatial data.

03 **REMOTE SENSING** — Collecting earth observation data from satellites and sensors.

04 **BIG GEOSPATIAL DATA** — Explosion of diverse, high-volume, high-velocity spatial data.

05 **MACHINE LEARNING** — Algorithms learn patterns from spatial data.

06 **DEEP LEARNING** — Deep neural networks extract complex spatial features.

07 **GeoAI** — AI + Spatial Intelligence for automated analysis, prediction and decision-making.

08 **DIGITAL TWIN & SMART CITIES** — Intelligent digital replicas and smart cities for a better, sustainable future.

- **Better Insights** — Understand complex patterns
- **Smarter Decisions** — Data-driven spatial intelligence
- **Greater Efficiency** — Automated analysis at scale
- **Sustainable Future** — Build resilient communities

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COMPONENTS OF GeoAI

GeoAI integrates diverse data sources, advanced AI methods, and actionable outputs to solve real-world spatial problems.



1 DATA SOURCES

- Satellite imagery
- UAV images
- LiDAR
- GNSS
- IoT sensors
- Mobile phones
- Social media
- OpenStreetMap

2 AI METHODS

- Machine Learning
- Deep Learning
- Reinforcement Learning
- Computer Vision
- Natural Language Processing

GeoAI

3 OUTPUTS

- Classification
- Prediction
- Optimization
- Pattern recognition
- Decision support

GeoAI turns spatial data into intelligence, empowering better decisions for a sustainable future.

- Understand Better
- Predict Accurately
- Optimize Resources
- Sustain Communities

GeoAI TECHNOLOGIES

Powering Intelligent Geospatial Solutions with Advanced AI, Machine Learning, Deep Learning and Spatial Analysis



WHY GEOAI TECHNOLOGIES?

- **Better Accuracy** — Learn complex spatial patterns.
- **Automation** — Reduce manual work and time.
- **Scalability** — Handle big geospatial data efficiently.
- **Smarter Decisions** — Support planning, management and policy.
- **Real-World Impact** — Enable smart cities, disaster management, environment monitoring, and more.

1 ARTIFICIAL INTELLIGENCE

- **Expert Systems** — Rule-based systems that mimic human expert knowledge.
- **Neural Networks** — Inspired by the human brain to learn complex patterns.
- **Reinforcement Learning** — Agents learn by interacting with the environment to maximize cumulative rewards.

2 MACHINE LEARNING

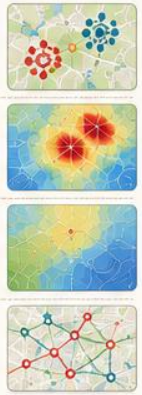
- **Random Forest** — Ensemble of decision trees for high accuracy.
- **Support Vector Machine (SVM)** — Finds the optimal boundary between classes.
- **XGBoost** — Gradient boosting framework for high performance.
- **Decision Trees** — Tree-structured models for interpretable decisions.

3 DEEP LEARNING

- **CNN** — Convolutional Neural Networks for image and spatial feature extraction.
- **RNN** — Recurrent Neural Networks for sequential and temporal data.
- **LSTM** — Long Short-Term Memory networks for capturing long-range dependencies.
- **Transformer Models** — Advanced attention-based models for complex spatial-temporal data.

4 SPATIAL ANALYSIS

- **Spatial clustering** — Identify groups or patterns in spatial data.
- **Hotspot analysis** — Detect statistically significant hot and cold spots.
- **Spatial interpolation** — Predict values in unsampled locations.
- **Network analysis** — Analyze connections and flows in spatial networks.



From Data to Intelligence

Collect → Process → Learn → Analyze → Decide → Impact

GeoAI Technologies Transform Spatial Data into Real-World Value

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WHY GeoAI MATTERS

GeoAI transforms spatial data into intelligent insights for a smarter and more sustainable future.

GeoAI supports



SMART CITIES
Optimize urban planning, resources, and services.

TRANSPORTATION
Improve traffic flow, route planning, and mobility solutions.

DISASTER MANAGEMENT
Predict, monitor, and respond to disasters effectively.

PUBLIC HEALTH
Detect disease patterns, monitor outbreaks, and support health planning.

AGRICULTURE
Monitor crops, predict yields, and optimize farming practices.

TELECOMMUNICATIONS
Optimize network planning, coverage, and performance.

CLIMATE CHANGE
Track environmental changes and support climate action.

ENVIRONMENTAL MONITORING
Monitor forests, water, air quality, and biodiversity.

BENEFITS

Faster analysis
Process large spatial datasets in minutes, not hours.

Improved accuracy
AI models reduce errors and enhance result reliability.

Real-time monitoring
Continuously track changes and detect events instantly.

Automated feature extraction
AI automatically identifies objects and patterns from spatial data.

Predictive modelling
Forecast future trends and support proactive decision-making.

GeoAI empowers better decisions, builds resilient communities, and creates a sustainable world.

Intelligent Insights | Data-Driven Solutions | Smarter Decisions | Resilient Communities | Sustainable Future

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GeoAI WORKFLOW

From Spatial Data to Intelligent Decisions

GeoAI integrates geospatial data and artificial intelligence to generate insights and support smarter decisions.



1 SPATIAL DATA COLLECTION



Collect data from multiple sources.

- Satellite imagery
- UAV / LiDAR
- GNSS
- IoT sensors
- Open data, APIs

2 DATA CLEANING



Improve data quality and consistency.

- Remove outliers
- Handle missing values
- Correct errors
- Standardize formats
- Ensure spatial accuracy

3 FEATURE ENGINEERING



Transform data into meaningful features.

- Spatial attributes
- Spectral indices
- Texture measures
- Proximity metrics
- Temporal features

4 AI MODEL TRAINING



Train AI models using prepared data.

- Select algorithms
- Train with datasets
- Tune parameters
- Optimize performance
- Prevent overfitting

5 MODEL VALIDATION



Evaluate model performance and reliability.

- Accuracy assessment
- Cross-validation
- Confusion matrix
- Performance metrics
- Robustness check

6 PREDICTION



Generate predictions on new spatial data.

- Classify
- Forecast
- Detect patterns
- Identify changes
- Estimate risk

7 DECISION SUPPORT



Convert predictions into actionable insights.

- Visualize results
- Generate reports
- Support planning
- Optimize resources
- Improve outcomes

VALUE DELIVERED

- Faster analysis
- Improved accuracy
- Real-time monitoring
- Automated feature extraction
- Predictive modelling
- Better decisions & impact

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GeoAI APPLICATIONS IN TELECOMMUNICATIONS

GeoAI combines geospatial data and artificial intelligence to power **smarter**, **faster** and more **efficient** telecom operations.



1 OPTIMIZE
TOWER PLACEMENT
Identify the best locations for new towers using AI and spatial analysis.

2 PREDICT
NETWORK DEMAND
Forecast user demand based on population, mobility and usage patterns.

3 DETECT
COVERAGE GAPS
Automatically detect weak or no-signal areas to improve service coverage.

4 IMPROVE
SIGNAL QUALITY
Analyze and enhance signal strength, interference and user experience.

5 PLAN 5G
DEPLOYMENT
Support strategic 5G rollout with AI-driven site selection and capacity planning.

6 MONITOR
INFRASTRUCTURE
Use GeoAI to monitor tower health, outages and asset performance in real time.

7 REDUCE
OPERATIONAL COSTS
Automate analysis and optimization to reduce site visits, energy use and downtime.

GeoAI
Smart Insights.
Stronger Networks.
Better Connections.

GeoAI IN ACTION

Data-Driven Decisions
Make smarter network decisions with real-time geospatial intelligence.

Better Customer Experience
Ensure strong, reliable connectivity where users need it most.

Future-Ready Networks
Build resilient and scalable networks for 4G, 5G and beyond.

Lower Costs
Optimize resources, improve efficiency and reduce operational expenses.

Sustainable Operations
Reduce energy consumption and support greener, smarter networks.

GeoAI
Turning Spatial Data into Smarter Networks

Geospatial Data → AI & Analytics → Network Intelligence → Better Connectivity → Business Impact

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GeoAI APPLICATIONS

NETWORK PLANNING

Using AI to determine

- Optimal cell tower locations
- Coverage optimization
- Capacity planning

INPUT DATA

- Population
- Land use
- Terrain
- Existing towers
- Traffic demand

AI-POWERED ANALYSIS

Spatial Data Integration → AI & Machine Learning Models → Predict & Evaluate Performance



OUTPUT

Optimal network design



- Better coverage
- Balanced capacity
- Cost-effective deployment
- Future-ready network

BENEFITS OF GeoAI

- Data-driven decisions
- Higher accuracy & efficiency
- Lower cost & risk
- Better coverage & capacity
- Improved user experience

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GeoAI APPLICATIONS

HUMAN MOBILITY ANALYSIS

Leveraging AI and geospatial data to understand how people move, why they move, and where they go.

DATA

- Mobile phone records
- GPS
- IoT
- Public transport



APPLICATIONS

Traffic prediction
Predict congestion, peak hours and travel times using mobility patterns.

Tourism analysis
Identify popular places, tourist flows and visitor behaviour.

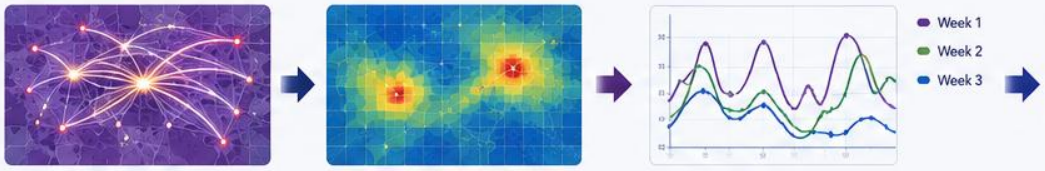
Population movement
Analyze migration, daily commuting and population distribution.

Pandemic monitoring
Track movement trends to support disease surveillance and control.

EXAMPLE

COVID-19 MOBILITY ANALYSIS

GeoAI uses mobile phone mobility data to monitor population movement during COVID-19 outbreaks.



BENEFITS OF GEOAI FOR HUMAN MOBILITY

- Real-time monitoring
- Accurate prediction
- Data-driven insights
- Efficient resource allocation
- Better planning & decision-making

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GeoAI APPLICATIONS

INFRASTRUCTURE MONITORING

GeoAI enables **intelligent** monitoring of telecommunication infrastructure for better **reliability** and **performance**.

MONITOR

- CELL TOWERS



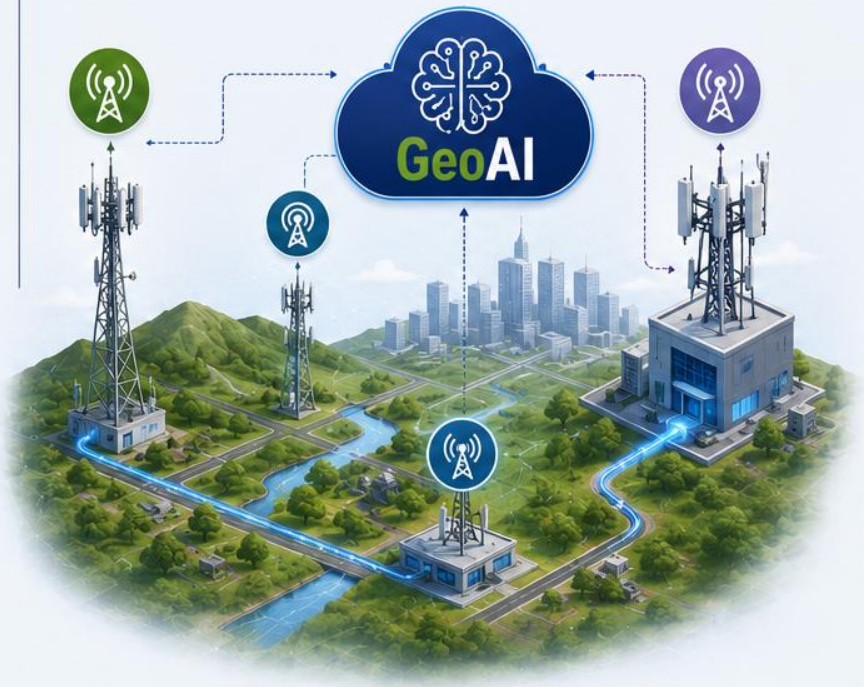
- FIBRE OPTIC NETWORK



- BASE STATIONS



SMARTER MONITORING. STRONGER NETWORKS.



USING

- SATELLITE IMAGERY



- UAV



- COMPUTER VISION



HOW IT WORKS

1. **DATA ACQUISITION** — Collect data from satellites, UAVs and network sensors.
2. **DATA PROCESSING** — Clean, integrate and prepare data for analysis.
3. **AI ANALYSIS** — Detect anomalies, damage and performance issues.
5. **ACTION & RESPONSE** — Take action, schedule maintenance and monitor improvements.

BENEFITS

- **EARLY FAULT DETECTION** — Detect faults and anomalies before they impact service.
- **MAINTENANCE PLANNING** — Plan proactive maintenance based on real-time insights.
- **REDUCED DOWNTIME** — Minimize service interruptions and improve network reliability.

GeoAI transforms data into actionable insights for resilient, efficient and future-ready networks.

AI-Powered Intelligence | Real-Time Monitoring | Better Decisions | Stronger Networks

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GeoAI APPLICATIONS

SMART CITIES

GeoAI combines geospatial data and artificial intelligence to build smarter, safer and more sustainable cities.

GeoAI supports

Intelligent transportation
Optimize traffic flow, reduce congestion and improve mobility.

Smart utilities
Monitor and optimize energy, water and waste systems.

Public safety
Predict risks, detect anomalies and enhance situational awareness.

Emergency response
Faster incident detection, resource allocation and disaster management.

Digital Twin
Create virtual replicas of cities for simulation, planning and decision-making.

Autonomous vehicles
Enable safe navigation and intelligent route planning.

Data + AI + Location Intelligence = Smarter Cities



HOW GEOAI POWERS SMART CITIES

DATA COLLECTION
Satellite, IoT, sensors, UAV, mobile, and open data

DATA INTEGRATION
Geospatial data integration and management

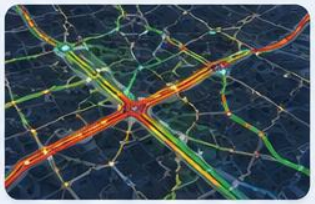
AI ANALYTICS
Machine learning & deep learning extract insights

DECISION SUPPORT
Real-time insights for planners, operators and citizens

SMART OUTCOMES
Better services, efficient operations and sustainable cities

EXAMPLE

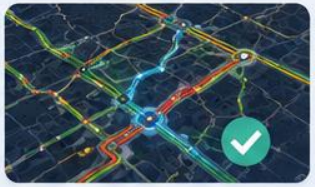
Real-time traffic optimization using AI.



Collect real-time traffic and mobility data



AI analyzes patterns and predicts congestion



Optimize traffic signals and suggest best routes in real time

BENEFITS OF GEOAI

- Data-driven decisions
- Real-time monitoring
- Improved efficiency & productivity
- Enhanced safety & resilience
- Sustainable communities

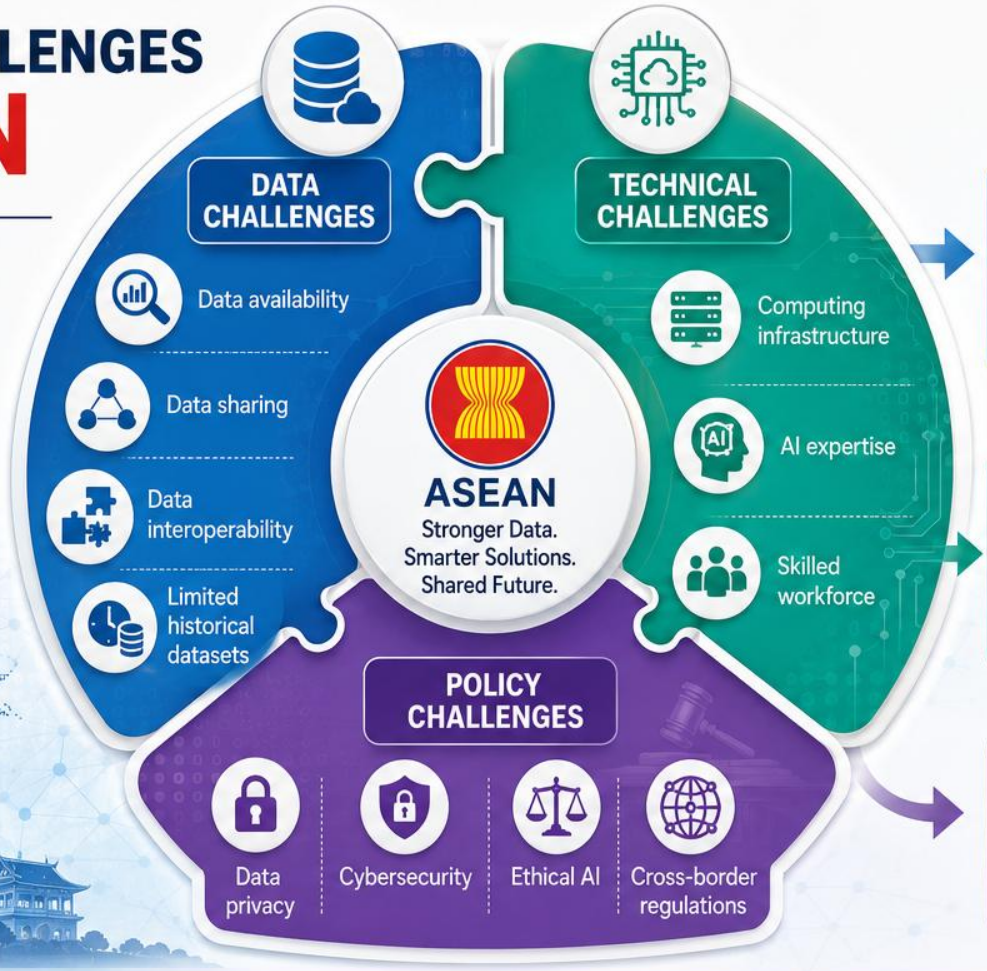
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REGIONAL CHALLENGES IN ASEAN

Realising the power of GeoAI across ASEAN requires overcoming key data, technical, and policy challenges together.







DATA CHALLENGES
Fragmented data, inconsistent standards, and limited access hinder regional insights and innovation.

TECHNICAL CHALLENGES
Building the right infrastructure, capabilities, and talent is essential to unlock the potential of GeoAI.

POLICY CHALLENGES
Trusted governance, ethical frameworks, and harmonised policies are critical for safe and inclusive collaboration.

COLLABORATION IS OUR COMPETITIVE ADVANTAGE

Share data responsibly | Build capacity together | Innovate for impact | Shape inclusive policies

A UNITED ASEAN, A SMARTER FUTURE.

OPPORTUNITIES IN ASEAN

Harnessing Technology, Innovation and Collaboration for a Sustainable Future



ASEAN INITIATIVES

ASEAN Smart Cities Network (ASCN)

Driving smart, sustainable and livable cities through collaboration and knowledge sharing.

ASEAN Digital Masterplan 2025

Accelerating the region's digital transformation for an inclusive and innovative digital economy.

ASEAN Sustainable Urbanisation Strategy

Promoting resilient, low-carbon and inclusive urban growth for a sustainable ASEAN.

Stronger Collaboration | Technology Innovation | Inclusive Growth | Sustainable Future

TOGETHER, BUILDING A RESILIENT AND PROSPEROUS ASEAN

CASE STUDY 1

SINGAPORE SMART NATION

Harnessing GeoAI to build
a smarter and more livable Singapore

GeoAI USED FOR

Traffic prediction
AI models analyse real-time traffic data to predict congestion and optimize traffic flow.

Public transport optimization
GeoAI helps optimize routes, schedules and capacity for efficient and reliable public transport.

Smart surveillance
AI-powered analytics and spatial data integration enhance public safety and threat detection.

Urban planning
GeoAI and spatial analysis support data-driven planning for sustainable and resilient urban development.



TECHNOLOGIES

IoT
Sensors and connected devices collect real-time data across the city.

AI
Machine learning and deep learning extract insights and make predictions.

GIS
Spatial data integration and analysis support better decision-making.

Digital Twin
Virtual models simulate the city to test scenarios and improve outcomes.

OUTCOME

IMPROVED MOBILITY AND PUBLIC SERVICES.

- Smarter decisions
- Greater efficiency
- Safer communities
- Sustainable urban growth

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CASE STUDY 2

INDONESIA
DISASTER MANAGEMENT

GeoAI for a Safer and More Resilient Indonesia

GEOAI INTEGRATES

SATELLITE IMAGERY
Real-time and historical satellite data

UAV
High-resolution aerial monitoring

RAINFALL
Rainfall intensity and distribution

DEM
Terrain elevation and slope analysis

POPULATION
Population density and vulnerability data



APPLICATIONS

FLOOD PREDICTION
Predict flood extent and impact using integrated hydro-environmental data.

LANDSLIDE MONITORING
Detect and monitor areas prone to landslides in near real-time.

EMERGENCY RESPONSE
Support decision-making for evacuation, resource allocation, and relief efforts.

OUTCOME

FASTER DISASTER RESPONSE.
Better preparedness, reduced losses, and safer communities.

Real-time insights | Accurate risk assessment | Data-driven decisions | Stronger coordination and communication | Resilient and sustainable future

CASE STUDY 3

MALAYSIA

Harnessing GeoAI for Smarter, Resilient and Sustainable Development

GEOAI APPLICATIONS

SMART CAMPUS
AI-powered campus management for energy, mobility and safety.

FLOOD MODELLING
Predict flood extent and risk using hydrological and spatial data.

LAND USE MAPPING
AI-driven classification and monitoring of land use changes.

FOREST MONITORING
Detect deforestation and forest health using remote sensing and AI.

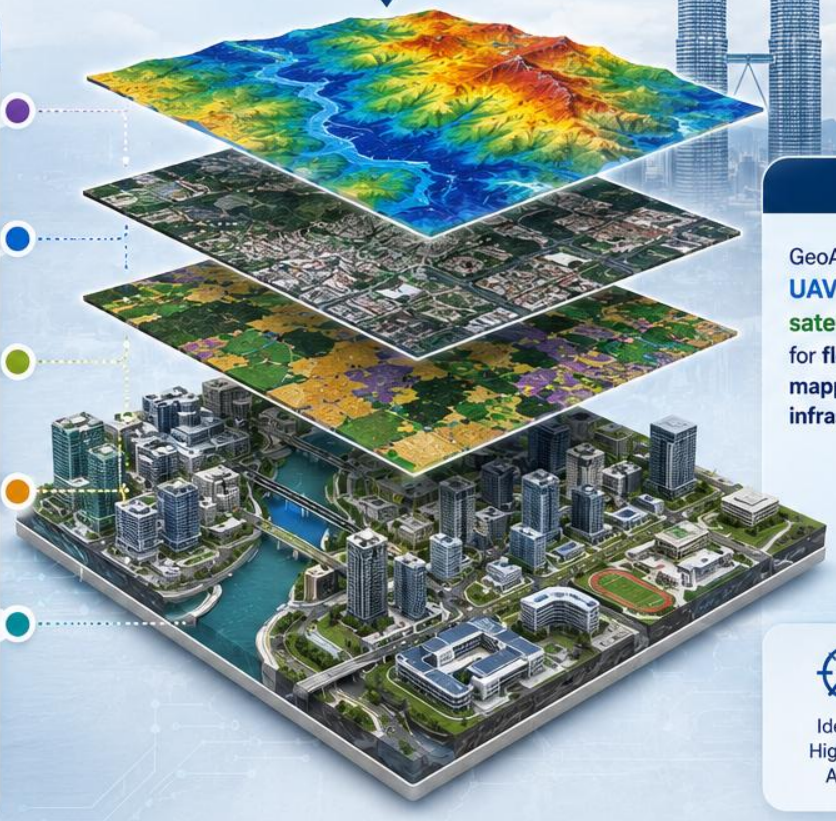
TELECOMMUNICATION PLANNING
Optimal site selection and network coverage analysis using GeoAI.

DIGITAL TWIN
3D virtual representation for urban simulation and decision-making.

GEOAI INTEGRATION

GEOSPATIAL DATA

AI / ML ANALYTICS







EXAMPLE

GeoAI integrated with **UAV** and **satellite imagery** for **flood susceptibility mapping** and **infrastructure planning.**



Identify High-Risk Areas

Plan Resilient Infrastructure

Support Data-Driven Decision Making

DELIVERING IMPACT

- Improved planning and governance
- Reduced disaster risk and losses
- Sustainable resource management
- Cost-efficiency and innovation
- Better quality of life

SUMMARY

GeoAI: Intelligent Insights. Smarter Decisions.

GeoAI

Where Intelligence Meets Location

GeoAI COMBINES

ARTIFICIAL INTELLIGENCE
Algorithms learn, predict, and automate decisions.

GIS
Spatial analysis, visualization, and decision support.

REMOTE SENSING
Earth observation from satellites, UAVs and sensors.

BIG GEOSPATIAL DATA
Large volume, diverse, real-time datasets drive deeper insights.



APPLICATIONS INCLUDE

TELECOMMUNICATIONS
Network planning, coverage analysis and infrastructure optimization.

SMART CITIES
Urban planning, smart utilities, asset management and liveability enhancement.

DISASTER MANAGEMENT
Risk assessment, early warning, impact analysis and response optimization.

TRANSPORTATION
Traffic prediction, route optimization, and mobility management.

CLIMATE MONITORING
Environmental monitoring, carbon tracking and climate resilience.

PUBLIC HEALTH
Disease mapping, outbreak prediction and health resource planning.

KEY MESSAGE

GeoAI transforms geospatial data into intelligent, predictive, and automated decision support systems.

Intelligent Insights | Predictive Analytics | Automated Workflows | Better Decisions | Stronger Communities